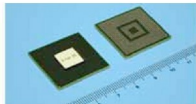


New Products

COMPONENTS

Automotive SoCs

Renesas Electronics Corporation has added R-Car H1 to its R-Car series of automotive system-on-chips (SoCs). This SoC is capable of delivering up to 11,650 dhystone MIPS (DMIPS).



The SoC is powered by Renesas' IMP-X3 core and Imagination Technologies' SGX-543-MP2 graphics processing unit (GPU), and delivers up to 40 gigaflops per second. It has 500MHz maximum operating frequency and 32-bit × two-channel data bus width.

The SoC is equipped with two independent DDR3 32-bit interfaces. Integrated peripherals include Ethernet, MOST-150, PATA and SATA interfaces, and PCIe for system expansion.

Renesas Electronics Singapore Pte Ltd, Bengaluru
Fax: (080) 67208777
Phone: 67208700
Web: www.renesas.com

32-bit microcontrollers

Microchip Technology Inc. has introduced a series of low-pin-count 32-bit microcontrollers that provide 61 DMIPS of performance in packages as small as 5x5 mm².



The PIC32 MX1 and MX2 are rated for operation at up to 105°C. These include up to 32 kB of Flash, 8 kB of static random-access memory, two PS interfaces for audio processing, Microchip's charge-time measurement unit peripheral for adding mTouch capacitive-touch buttons or advanced sensors, and an 8-bit parallel master port interface for graphics or external memory.

The microcontrollers also feature an on-chip 10-bit, 1-mega-samples-per-second, 13-channel analogue-to-digital converter, as well as USB 2.0 and serial-communication peripherals. Microchip Technology (India) Pvt Ltd, New Delhi
Fax: (011) 51608632
Phone: 41608631
Web: www.microchipdirect.com

Multi-mode transceiver chip

Fujitsu Semiconductor has released a multi-mode transceiver chip. The MB86L12A can eliminate external low-noise amplifiers and inter-stage surface-acoustic filters from the transmit and receive paths of both 3G and LTE line-ups.

The receiver also incorporates anti-aliasing filters, digital channel filters, digital gain control and high-dynamic-range analogue-to-digital converters. It supports GSM (GSM850, EGSM900, DCS1800 and PCS1900), WCDMA (bands I, II, III, IV, V, VI, VIII, IX, X and XI) and LTE (FDD bands 1, 3, 4, 6, 7, 8, 9, 10, 11, 13 and 17, and TDD bands 38 and 40).

Fujitsu Semiconductor Asia Pte Ltd, Bengaluru
Fax: (080) 28416660
Phone: 28419990
Web: www.fujitsu.com

14-bit analogue-to-digital converter

Texas Instruments Inc. has introduced a 14-bit analogue-to-digital converter (ADC) that delivers 78.2dBFS signal-to-noise ratio at 5 MHz and 84dBc spurious-free dynamic range at 5 MHz/80 MSFS.

The ADS5294 offers 77 mW per channel at 80 MSFS (two low-voltage differential signalling wires per channel)



and low-frequency noise-suppression mode. It also has a digital processing block that integrates system functions such as two- or four-channel averaging and decimation by two, four or eight filters.

Texas Instruments India, Bengaluru
Phone: (080) 41381665
E-mail: tiasia@ti.com
Web: www.ti.com

Low-noise op-amp

Maxim Integrated Products has introduced a 20V, precision, low-noise dual operational amplifier (op-amp). The MAX44251 employs auto-zero technology that enables continuous self-calibration.



tion. Auto-zeroing constantly measures and compensates the input offset, thus eliminating drift over time and temperature and the effect of 1/f noise.

The op-amp consumes only 1.15 mA per channel, while providing 10MHz gain bandwidth and 5.9nV/√Hz input-referred voltage noise. Its power-supply rejection ratio, common-mode rejection ratio and large-signal voltage gain exceed 135 dB at +25°C. The MAX44521 is rated for -40°C to +125°C temperature